

ELYON-SOL WHITEPAPER (v0.9.8.4 CANONICAL)

Governance Before Intelligence

A Pre-Execution Substrate for Deterministic Refusal, Authority Validation, and Continuity Preservation

Abstract

Modern AI and policy-driven systems prioritize correctness, performance, and compliance, yet they often overlook a foundational question: *Should this interaction occur at all?*

Elyon-Sol is a governance-first substrate operating prior to policy evaluation or execution. It enforces deterministic refusal based on authority (AC^3), coverage (T^{26}), and continuity (CCS), with failure constructs including CDD, SAP, PAD, and ILT.

Version **v0.9.8.4** resolves specification consistency, finalizes structural clarity, and aligns evaluation logic across all sections. No new invariants or behaviors are introduced. Elyon-Sol remains a deterministic governance specification for pre-execution admissibility, not a complete formal proof system.

1. Introduction

AI systems now operate in environments where unauthorized or illegitimate interactions carry significant risk. Traditional approaches assume an interaction should proceed to evaluation. Elyon-Sol instead evaluates whether an interaction should exist at all.

2. System Overview

Elyon-Sol is a non-executing governance substrate.

Evaluation Pipeline:

- Authority (AC^3)
- Coverage (T^{26})
- Continuity (CCS)

Outcome:

- ELIGIBLE

- REFUSE

3. Governance Invariants

AC³ — Authority Construct

All required authority must be present, identifiable, and properly scoped.

Authority includes required consent conditions where applicable.

T²⁶ — Coverage Model

All required participants, roles, and evidence must be present.

CCS — Continuity Control Surface

System identity, structure, and semantics must remain consistent across transitions.

Evaluation Rule

Failure of any invariant results in immediate refusal.

Notation Clarification

AC³ denotes authority completeness; T²⁶ denotes coverage completeness.

AC³ and T²⁶ are nominal designations from original Elyon-Sol notation; the superscripts are not mathematical exponents.

4. Interaction-Level Failure Constructs

Elyon-Sol defines four detection-layer constructs describing failure conditions:

CDD — Consent Deadlock Detection

Occurs when a required consent condition exists but cannot be satisfied.

SAP — Service Access Paradox

Occurs when a system requires a condition for access that it simultaneously prevents.

PAD — Policy Authority Disclosure

Occurs when a system exposes its constraints without providing a valid resolution path.

ILT — Illegitimate Loop Termination

Occurs when a system continues interaction despite absence of any valid or legitimate path forward.

These constructs describe failure modes but do not participate in admissibility determination.

See Section 15 for full operational definitions.

5. Case Studies (Abstracted)

Elyon-Sol evaluation behavior is demonstrated through worked examples in Appendix D.

Observed patterns include:

- consent denial with continued interaction
- coverage failure with continued processing
- termination without resolution

All cases resolve deterministically under AC^3 , T^{26} , and CCS.

6. Lightweight Formal Model

```
def evaluate(ctx):
```

```
    if not ac3_valid(ctx): return REFUSE
```

```
    if not t26_valid(ctx): return REFUSE
```

```
    if not ccs_valid(ctx): return REFUSE
```

```
    return ELIGIBLE
```

Scope Clarification

This document defines a deterministic specification of interaction admissibility. It provides formal definitions and operational constraints sufficient for implementation but does not claim a complete mathematical proof system or empirical validation across all domains.

7. Regulatory Alignment (EU AI Act)

Elyon-Sol operates as an upstream admissibility gate that can support compliance with:

- Article 5 (prohibited practices) by preventing illegitimate interactions
- Article 9 (risk management) through fail-closed evaluation
- Article 10 (data governance) via explicit coverage requirements
- Article 14 (human oversight) by enforcing authority validation

Elyon-Sol does not replace compliance systems but constrains interaction eligibility prior to their application.

8. Operational Integrity

8.1 Manifest-Bound Authority

Governance is bound to a deterministic manifest defining required authority and coverage.

8.2 Proof-of-Existence (PoE)

PoE anchoring provides an optional mechanism for verifying artifact integrity. The choice of anchoring system is implementation-dependent and does not affect admissibility logic.

8.4 Implementation Patterns (Non-Canonical)

GAE/ARL are implementation patterns derived from the canonical evaluation function. They do not introduce new evaluation criteria.

9. Reproducibility

A valid Elyon-Sol implementation must:

- deterministically derive $AR(I)$ and $R(I)$ from the governing manifest M
- produce identical evaluation results for identical inputs under the same manifest
- fail closed under any missing or invalid input conditions

Reproducibility is defined at the evaluation level, not at the reasoning layer.

10. Limitations and Scope

Elyon-Sol defines a deterministic specification of interaction admissibility.

It does not:

- provide a complete formal proof system over all domains
- perform probabilistic reasoning or inference
- execute actions or enforce policies directly

All guarantees are bounded to invariant evaluation and fail-closed behavior.

11. Formal Interaction Model

11.1 Interaction

$I = (A, S, C, t)$

A = actors

S = system state

C = context

t = time

11.2 Actors

Identity $\neq$ authority

11.3 Required Authorities

$AR(I)$

11.4 Coverage Requirements

$R(I)$

11.5 Present Authorities

$AP(I)$

11.6 Observed Coverage

$OP(I)$

11.7 Authority Evaluation

$AC^3(I) = 1 \Leftrightarrow AP(I) \supseteq AR(I)$

11.8 Coverage Evaluation

$T^{26}(I) = 1 \Leftrightarrow OP(I) \supseteq R(I)$

11.9 Governing Manifest

$M: I \rightarrow (AR(I), R(I))$

All required authority and coverage sets are derived exclusively from M and evaluated without inference or substitution.

The manifest must be deterministic, versioned, and integrity-verifiable.

12. Continuity (CCS)

12.1 State Transition

$S_t \rightarrow S_{t+1}$

A transition occurs on any change in interaction context, authority, coverage, or system state.

12.2 Decision Variables

$u = AC^3(I)$

$c = T^{26}(I)$

$d = u \wedge c$

The decision state d represents the admissibility condition derived from authority and coverage evaluation.

12.3 Continuity Constraint

$CCS(S_t, S_{t+1}, I) = 1$ iff:

- authority transitions are justified by $AC^3(I)$
- coverage transitions are justified by $T^{26}(I)$
- decision consistency holds:
 $d_{t+1} = u_{t+1} \wedge c_{t+1}$

Continuity requires that authority, coverage, and decision semantics remain internally consistent across transitions or be revalidated.

12.4 Failure Condition

If any condition is violated:

CCS = 0

Any transition that alters authority, coverage, or decision state without valid re-evaluation constitutes a continuity violation.

Examples of invalid transitions include:

- governing manifest version change
 - role or authority schema change
 - identity or mapping inconsistency
-

13. Evaluation Function

$G(I) = AC^3(I) \wedge T^{26}(I) \wedge CCS(S_t, S_{t+1}, I)$

If $G(I) = 1 \rightarrow \text{ELIGIBLE}$

If $G(I) = 0 \rightarrow \text{REFUSE}$

Eligibility does not persist across state transitions without revalidation.

14. Scope Clarification

Elyon-Sol:

- governs legitimacy
- operates pre-execution
- is identity-agnostic

Does NOT:

- execute actions
 - replace identity systems
 - function as a policy engine
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15. Interaction Failure Constructs (Operational Definitions)

CDD — Consent Deadlock Detection

A required consent condition exists but cannot be satisfied.

No valid interaction path remains.

SAP — Service Access Paradox

A system requires a condition that it prevents from being satisfied.

PAD — Policy Authority Disclosure

Constraints are disclosed without a valid resolution path.

ILT — Illegitimate Loop Termination

Interaction continues despite absence of any valid or legitimate path forward.

These constructs describe failure modes and do not alter admissibility logic.

Appendix D — Clarifications and Demonstrations

D.2 Positive Case (True ELIGIBLE Outcome)

Interaction Context (Anonymized)

Actor: authenticated user

Role: authorized operator

Artifacts: valid session, request payload

Evaluation

$AP = \{\text{authenticated identity, valid role}\}$

$AR = \{\text{authenticated identity, valid role}\}$

$\rightarrow AC^3(I) = 1$

$OP = \{\text{request context, valid session}\}$

$R = \{\text{request context, valid session}\}$

$\rightarrow T^{26}(I) = 1$

No state transition inconsistency observed

$\rightarrow CCS = 1$

Result

$G(I) = 1 \wedge 1 \wedge 1 = 1 \rightarrow \text{ELIGIBLE}$

D.3 CCS-Isolated Failure Case

Interaction Context (Anonymized)

Actor: authenticated user

Role: authorized operator

Artifacts: valid session, request payload

Evaluation

AP = {authenticated identity, valid role}

AR = {authenticated identity, valid role}

→ $AC^3(I) = 1$

OP = {request context, valid session}

R = {request context, valid session}

→ $T^{26}(I) = 1$

State transition introduces inconsistency in interpretation or mapping

→ $CCS = 0$

Result

$G(I) = 1 \wedge 1 \wedge 0 = 0 \rightarrow \text{REFUSE}$

D.4 Relation to Prior Work

RBAC / ABAC / XACML: assume interaction legitimacy

UCON: governs usage but not existence

Reference Monitor: mediates access but not admissibility

Elyon-Sol evaluates whether an interaction should exist at all.

Canonical Constraints

- No new invariants introduced
 - No probabilistic evaluation
 - Fail-closed enforced
 - No execution-layer expansion
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Final Statement

ELIGIBLE

Otherwise: REFUSE